

Cognitive-Behavioral Therapy for Children and Adolescents

*Danielle Cornacchio, Amanda L. Sanchez,
Tommy Chou and Jonathan S. Comer*

Florida International University, Miami, FL, United States

Children's mental health problems impose a staggering public health burden. Roughly half of the population suffers from a mental disorder at some point in their lifetime (Kessler et al., 2005), with the majority of these problems onsetting during childhood or adolescence. In fact, somewhere between 13% and 40% of youth meet criteria for a mental disorder within the past 12 months (Costello, Egger, & Angold, 2005; Kessler et al., 2012; Merikangas et al., 2010), and without intervention these conditions do not typically remit over time. These estimates are particularly concerning when considering the tremendous individual, familial, and societal costs associated with children's mental health problems, including family dysfunction, peer problems, academic impairments, physical health problems, and criminality, as well as overall reduced quality of life, poorer worker productivity, and increased health care costs across the lifespan (e.g., Comer et al., 2011; Kessler et al., 2012). Indeed, effective intervention for mental health problems presenting in childhood is critical.

Despite daunting statistics regarding the prevalence, scope, and impact of children's mental health problems, the past few decades have witnessed tremendous advances in the development and evaluation of evidence-based psychological procedures with demonstrated success in treating a considerable share of children's mental health problems. These supported practices have largely drawn on developmentally informed cognitive-behavioral models of intervention and have shown

compelling efficacy across hundreds of controlled studies evaluating indicated treatment methods for children's mental health problems.

Given this progress, in this chapter we provide an overview of the cognitive-behavioral treatment (CBT) model for mental health problems, followed by a discussion of key developmental considerations and distinguishing attributes of CBT when targeting mental health problems in youth. We then turn our attention to how CBT differentially addresses specific domains of child mental health problems (i.e., anxiety, depression, trauma, attention-deficit/hyperactivity disorder (ADHD), and disruptive behavior problems), and then we review major advances and innovations in recent years that hold great promise for meaningfully expanding the reach and quality of CBT for an even a larger proportion of affected children and adolescents. These innovations include the development of transdiagnostic, modular, and intensive CBT formats for child problems, as well as the promising role of technology for improving the accessibility of CBT for traditionally underserved child populations. We conclude by highlighting areas in need of continued empirical attention and laying out an agenda for future research seeking to optimize the public health impact of CBT procedures for redressing children's mental health problems.

THE CBT MODEL OF PSYCHOLOGICAL TREATMENT

Broadly speaking, CBT refers to a family of treatments that target psychopathology by focusing on problematic associations between thoughts, behaviors, and emotions maintained by behavioral (e.g., reinforcement, avoidance) and/or cognitive (e.g., memory, attribution, attention) processes (Boswell et al., 2011; Hofmann, 2011). CBT can be thought of as the systematic integration of behavioral methods of conditioning and reinforcement that grew out of the earlier work of Pavlov (1927), Mowrer (1939), Skinner (1953), Watson and Raynor (1920), and Wolpe (1952) with a focus on cognitive distortions, attributions, and related mental phenomena popularized in the 1960s and 1970s by Ellis (1962) and Beck (1976). In recent decades, findings from affective science have increasingly informed CBT models, leading to emotion and its dysregulation taking a more central role in modern CBT (e.g., Hofmann, Sawyer, Fang, & Asnaani, 2012; Power & Dalgleish, 2008; Samoilov & Goldfried, 2000).

CBT has long been distinguished from more traditional models of psychotherapy by its present-focused and data-driven approach. CBT therapists efficiently focus on symptom-maintaining factors in the "here and now" rather than search for historical antecedents that may or may not be related to present psychopathology. Moreover, CBT is

goal-focused and action-oriented, and for these reasons typically time-limited. CBT therapists collaborate with their clients to identify specific and obtainable goals, and together CBT therapists and their clients routinely monitor progress toward these goals. CBT therapists are typically more directive than more traditional therapists. For example, whereas psychoanalysts might prioritize free association and play therapists might prioritize unstructured play in treatment, CBT therapists take an active role in shaping the structure and content of sessions, creating session agendas, and working to actively build new skills and more adaptive coping repertoires in their clients. Moreover, CBT therapists typically assign out-of-session homework in order to facilitate generalization of treatment to real-world situations. Indeed, research indicates that homework completion outside of treatment is an important and significant predictor of treatment success (Kazantzis, Deane, & Ronan, 2000).

EIGHT KEY COMPONENTS OF CBT PRACTICE ACROSS THE LIFESPAN

Before introducing special considerations and distinguishing characteristics of CBT when working with child populations, it is important to first provide a general overview of the key components found across CBT practices for problems regardless of client age. Broadly speaking, CBT protocols for psychological problems across the lifespan typically entail some combination of eight key components of CBT practice. Although we go into further depth regarding many of these components later in this chapter when we discuss focused CBTs for specific child problems, we first briefly introduce these components here as part of our CBT overview.

The first key CBT component is providing *psychoeducation* to clients about the problems that have brought them into treatment. Clients are provided with information about their mental health conditions so that they can better understand, accept, and destigmatize their problems, recognize that they are not alone in their suffering, and accept taking on an active role in making indicated changes. Second, as noted, CBT is a data-driven approach to treatment, and as such *routine symptom monitoring* is a critical component of CBT care. Mood, thoughts, and behaviors are routinely assessed in CBT at regularly fixed intervals and provide data that in turn continually inform the course of treatment over time—including necessary treatment adjustments and course corrections. In addition to informing the direction of treatment, having clients engage in monitoring of their own moods, thoughts, and behaviors can raise their awareness of maladaptive

patterns and maintaining influences and produce therapeutic effects. Third, *somatic management* is often incorporated into CBT to reduce tension in the body and can include diaphragmatic breathing exercises and progressive muscle relaxation.

The fourth CBT component is *contingency management*, which entails the use of interpersonal or tangible rewards to reinforce positive behaviors, and sometimes the incorporation of response costs for undesired behaviors. *Behavioral engagement* is the fifth key CBT component and varies depending on the specific problem being treated. For example, for depression, behavioral engagement often takes the form of activity scheduling and behavioral activation, whereas for anxiety, behavioral engagement might take the form of graduated exposure to the situations and stimuli that the client fears or avoids. For many problems, behavioral engagement will mean rehearsal of key skills newly acquired in treatment, such as social skills, assertiveness, anger management, effective communication, distress tolerance, healthy eating, or self-care.

Sixth, CBT often engages clients in *cognitive restructuring and related thought challenging exercises* in which clients learn to identify and adjust maladaptive thoughts and distortions that maintain psychopathology and negatively affect quality of life. Seventh, *problem-solving* entails helping clients take an active role in defining their problems, identifying possible solutions, evaluating the pros and cons of various solutions, determining which solutions are most likely to result in positive outcomes, enacting selected solutions, and then monitoring the effectiveness of enacted solutions. Finally, *relapse prevention* focuses on consolidating skills learned in treatment, with an emphasis on maintaining gains after treatment terminates, preventing the occurrence of symptom or behavioral lapses going forward, identifying early warning signs that problems may be returning, and developing an action plan for lapse-management strategies.

DISTINGUISHING ATTRIBUTES AND KEY CONSIDERATIONS FOR CBT WHEN WORKING WITH CHILD POPULATIONS

Although the underlying theory, science, principles, and components of CBT remain the same regardless of client age, a number of distinguishing attributes and key considerations are specific to the treatment of children and adolescents, and merit focused consideration. These include considerations of (1) developmental compatibility, (2) therapeutic alliance and the posture of the therapist, and (3) therapy participants and settings.

Developmental Compatibility

Many of the CBT strategies that have shown great success in adult populations rely heavily on methods and tasks that are beyond the developmental capacities of younger children and adolescents, and so simple downward extensions of treatments designed for older individuals to address the clinical needs of youth are misguided (Carpenter, Puliafico, Kurtz, Pincus, & Comer, 2014; Grave & Blissett, 2004; Kingery et al., 2006). Importantly, supported CBT for adults requires a basic understanding of causality and of mental states and requires the individual to engage in representational thought and systematic problem-solving (Weisz & Weiss, 1989). Although common strategies such as the use of pictorial representations have been suggested as ways to increase the “digestibility” of content in light of linguistic challenges (Kingery et al., 2006), deficits in reasoning and other advanced cognitive functions still impinge on children’s capacity to engage in content focusing on maladaptive or inaccurate cognition and perception (Grave & Blissett, 2004).

Consider the following clinical misstep from a CBT trainee working with a 5-year old child with social anxiety disorder:

Therapist: *What if you were to have thought “the other kids are not going to laugh at me”? Do you think you would’ve still felt scared?*

Child: *(Blank stare from child)*

Therapist: *Do you think in the future you could try to think “the other kids aren’t going to laugh at me”*

Child: *(Blank stare from child)*

This very brief vignette illustrates a great number of ways that a CBT therapist can fail to take into account a younger child’s restricted but expanding cognitive development. We now consider five overlapping domains of developing cognition in youth as they pertain to this brief vignette and how they pertain to successful engagement in various CBT tasks.

Mental Representation

Mental representation refers to the ability to go beyond what is observable and immediately available to the senses (Flavell, Miller, & Miller, 2001). One of the very earliest accomplishments in this area is found in the infant’s emerging appreciation for object permanence, in which they develop an awareness that objects and people exist independent of their physical perceptions of them. By the end of the first year of life, if you cover a toy with a blanket in front of the child, the child will likely pick up the blanket to find the toy. Although typically developing children develop a sophisticated appreciation for object permanence by

the end of their first year, increasingly sophisticated capacities for mental representation continue to unfold deep into young adulthood. Language development is a critical accomplishment in this arena, as children progressively learn that sounds, words, and then clusters of words can have increasingly nuanced representational meanings. Imagery then entails going beyond simple mental representations and zooming in on the sights, sounds, smells, tastes, and textures of mental constructions of the world. Although there's evidence of basic imagery in infancy, this process becomes more and more refined as children develop—and even into adulthood we are still developing increasing abilities for detail and vividness in our imagery.

With regard to CBT for youth, any in-office cognitive approach directly working with children requires the child to have a basic ability to mentally represent objects, persons, situations, and scenarios. When asking children to consider anything *outside* of the office—whether it is consideration of the home, family life, school, or peer scenarios—the child essentially has to be able to mentally leave the office. For example, in the misguided therapy vignette above, the therapist had the child consider his classroom situation while in the therapy office, but it is not clear how well the child was actually able to mentally represent his classroom or to recreate that specific day while in the therapy office and while in the presence of so much novel perceptual stimuli.

Even after children have acquired a basic ability to mentally represent, and they are able to engage broadly in various forms of cognitive therapy, more sophisticated representational abilities are required for many conventional CBT techniques originally developed for use with adults. For example, imaginal exposures for anxiety have the client imagine in detail certain aspects of feared objects or situations, with the assumption that repeatedly encountering feared material, even in imaginal forms, can extinguish feared associations. If a child is unable to achieve vivid images, he or she may be unable to imagine feared objects or situations in sufficient detail to allow for meaningful deconditioning to actually occur.

It is also critical to consider the child's developing ability to mentally represent the self. When asked to describe themselves, preschoolers will tend to focus on physical characteristics (e.g., that they are boy or girl) and then self-representations progress to actions (e.g., defining themselves as someone who walks to school or who plays baseball). In middle childhood, children begin to define themselves in more psychological terms (e.g., that they are a child who *likes* to play baseball), and then in the context of other children (e.g., that they are someone who plays baseball *better* than other kids). Children's self-representations become more differentiated and nuanced with age (Flavell et al., 2001) and so, for example, an adolescent may be able to

understand himself as better than *most* kids at *baseball* but *not* as good as other kids at *math* or *art*. Given that cognitive work with youth suffering from depression often targets children's representations of the self, including self-esteem and self-worth, it is important that care is taken by the therapist to ensure that the encouraged new attitudes are consistent with the child's developing capacity to represent him or herself. For example, if a therapist were working with a young child who is feeling bad about her soccer abilities, it might be ineffective if the therapist were to encourage the child to consider that she is still better at soccer than a lot of kids, that she does not even like playing soccer, and that she is also really great at baseball.

Causal Reasoning

Causal reasoning refers to the ability to recognize the causes of events with the effects they produce. As early as 6 months, infants have already developed at least a rudimentary understanding of cause-and-effect relationships in the physical world, and by 10 months children can appreciate cause-and-effect relationships that cross sensory modalities, such as sight and sound (e.g., learning that when one squeezes a rubber duck, it produces a squeak) (Flavell et al., 2001). As children mature, they develop an improved capacity to comprehend temporal relationships among increasingly distal events and can begin to appreciate complex causal chains. But a more sophisticated understanding of causal relationships among thoughts, feelings, and behavior may not emerge until late adolescence and young adulthood (Flavell et al., 2001). For a young child who has not yet developed an ability to recognize cause-and-effect relationships between the mind and behavior, it might be misguided to attempt to engage the child in cognitive treatment tasks that emphasize the associations between thoughts, feelings, situations, and behavior. For example, when treating a young child with depression, it may be more appropriate to target behavioral activation and increasing positive reinforcement in the child's life, rather than working to convey to the child how thoughts can influence behavior and mood.

Counterfactual Reflection

Counterfactual reflection refers to the ability to consider what we know about the past and present in the context of what could have also been (Flavell et al., 2001). Essentially, it is the emergence of an ability to entertain *alternatives to reality*—to recognize what has happened or what is happening, and to simultaneously conceive of alternatives that could have happened instead. At around the age of four, children's language begins to reflect a rudimentary acknowledgment of realistic counterfactuals, but as late as age 7 children still show great difficulty entertaining

multiple possibilities in reasoning tasks that require consideration of counterfactuals (e.g., asking a child “what else could have happened?”). It then takes several more years before children are able to go beyond simply identifying alternatives and to report *why* alternatives are possible.

In the misguided clinical vignette above, when the therapist asked the 5-year-old child to consider how he would have felt if he had thought to himself that “maybe the other kids wouldn’t laugh at me,” not only was the therapist asking him to consider a causal relationship between thought and behavior, but the therapist was also asking him to imagine an alternative reality in which he thought and behaved differently from how things unfolded in reality.

Consideration of counterfactual reflection abilities is extremely important even as early as the first session, when the therapist considers what to explain to children about what therapy is. Explaining to a child that therapy is an activity that helps kids *change* things about themselves—whether the goal is to become a happier kid, or to be less anxious, or to make any other changes—requires the child to consider how they currently are and to appreciate that there are *alternatives* (e.g., appreciating that “I am sad, but it is a possibility that I could instead be happy”). Given younger children’s limited capacity for counterfactual reasoning, when working with very young children it can be most important to simply convey that the therapy is a time for the child to enjoy him or herself and have fun, rather than a confusing time, which could potentially offset the alliance by communicating a notion that is beyond their comprehension (i.e., that the therapist and child are going to work together to try to change some things about the way the child is).

Consideration of Future States and Hypotheticals

The ability to predict future states and hypotheticals refers to the ability to go beyond what *is* happening and to consider what *could* happen. By the age of three, children begin to offer future-oriented language, and soon after there is evidence of future-oriented behavior (e.g., planning, anticipation, and delay of gratification; [Flavell et al., 2001](#)). But young children still have great difficulty considering future states in the context of *uncertainty*—they might be able to consider what *will* happen but might have more difficulty considering what *could* happen. For example, a young child might understand that if you throw a ball up, it will come down, but it may be more difficult for the child to anticipate what their mother or father could do next. In the misguided clinical vignette above, when the therapist asked the 5-year old child “Do you think in the future you could try to think that other kids might not laugh at you?,” the therapist was asking the child to engage in a sort of “mental time travel” that may have been beyond his abilities.

Among the activities that require an ability to consider future states, let us consider problem-solving and planning, both of which are often key components of CBT protocols for older clients. Of course, in some cases, the child might have a somewhat delayed ability for problem-solving and planning, and this is the very target of treatment (e.g., in organizational skills training for kids with ADHD; Abikoff et al., 2013). But, particularly in earlier childhood, a limited capacity to engage in formal problem-solving and planning is not the treatment target itself; rather, limitations associated with typical development simply present obstacles to engaging children in treatment tasks. Accordingly, with younger children, CBT therapists have to focus less on the Socratic dialogues therapists use when engaging adults in problem-solving and planning and, instead, must rely more heavily on providing concrete practical skills, multiple opportunities for practice, and contingent reinforcement for positive behaviors.

Theory of Mind

Building on all of these cognitive abilities is theory of mind, which refers to the child's ability to develop a basic understanding and appreciation of the mind, an awareness of mental states and processes, and the ability to attribute mental states to others—particularly when these mental states *differ* from the child's *own* mental states (Flavell et al., 2001; Flavell, Green, & Flavell, 1998, 2000). Theory of mind considers the child's ability to recognize that other people's behavior can be attributed to their mental states (e.g., their thoughts, beliefs, desires, intentions, and knowledge) and that by attending to others' mental states the child can make the social world more predictable. These abilities are closely linked to perspective-taking, which involves being able to coordinate multiple points of view.

Children with autism and other pervasive developmental disorders struggle with tasks requiring a theory of mind (e.g., when being asked to predict the behavior of someone who does not have the same information and knowledge about the situation as the child). But, in typically developing children, theory of mind emerges rapidly, with even toddlers talking about actions in terms of mental states (Flavell et al., 1998, 2000). By the age of 4 or 5, children have developed a basic understanding of beliefs, desires, intentions, and visual perspectives; but, for the next several years children still struggle to develop an ability to engage in more sophisticated abstract reasoning tasks that require more than two perspectives at once or that require consideration of the role of mental states in hypothetical scenarios. Well into adolescence, typically developing teens are still developing abilities to coordinate multiple coexisting and seemingly competing perspectives (Devine & Hughes, 2013; Grazzani & Ornaghoi, 2012).

In the misguided clinical vignette above, when the therapist asked the 5-year-old child whether he could have considered in the past or in the future that other children would not laugh at him, the therapist was not only attempting to engage the child in mental time travel that was likely beyond his abilities, but the therapist was also asking the child to attribute others' thoughts and emotions and to link these mental states to their behavior in the context of hypothetical scenarios. According to the developmental literature (Flavell et al., 2001), this complicated task is considerably beyond a 5-year-old child's cognitive capacities. Similarly, in cognitive restructuring tasks, CBT therapists often use Socratic methods to have adults consider *how else* a given situation could be construed or perceived and how *other* people might consider a situation *differently*. These methods would require a child to consider multiple perspectives and their differential relationships to mood. As such, when working with younger children, it is best to stay away from tasks that emphasize the relativism and pluralism of perspectives. Instead, therapists should more concretely teach children optimistic perspectives, rather than engaging them to consider how particular outlooks might be associated more positive or negative mood states. However, research has shown that theory of mind training can result in meaningful improvements in theory of mind related skills when working with children with pervasive developmental delays (Hofmann et al., 2016).

Emotion Understanding and Regulation

Kingery et al. (2006) have considered emotional understanding and emotional regulation in the context of CBT for youth. Emotional understanding refers to identifying and appropriately labeling emotional experiences, whereas emotional regulation refers to managing ones emotions given fluctuating situational contexts and environmental demands. Emotional understanding develops throughout early childhood and shows dramatic growth through middle childhood and adolescence. Similarly, emotion regulation abilities advance in early childhood and into middle childhood and continue to gain sophistication in adolescence. Assessment of a child's emotional understanding and regulation capacity can play a major role in shaping the delivery of CBT.

Youth with deficits in emotional understanding may benefit from exercises working to improve the identification of emotions and their explanatory nature and relationship to thoughts and behaviors (Kingery et al., 2006; Suveg, Kendall, Comer, & Robin, 2006). Whereas younger children may have difficulty fully elaborating and grasping concepts related to emotions, adolescents might more readily adopt and cultivate an understanding of their own and others' thoughts and emotions as well as more complex motivations (Kingery et al.,

2006; Southam-Gerow & Kendall, 2000). Moreover, although activities such as labeling emotions of popular icons or cartoon characters may be the upper limit to which younger children are able to participate, adolescents may be able to take advantage of more detailed discussion and didactic presentation of emotion- and cognition-related content (Southam-Gerow & Kendall, 2000). Understanding these limitations in early and middle childhood can also prove essential when considering whether to shift to more active, practice-based content versus spending time in didactic sessions that may offer little benefit to the child (Kingery et al., 2006; Southam-Gerow & Kendall, 2000).

Therapeutic Alliance and the Posture of the Therapist

Therapeutic alliance is a broad construct that refers to the affective bond between the therapist and client and the agreement and collaboration between the therapist and client on therapy tasks and goals (DiGiuseppe, Linscott, & Jilton, 1996). Therapeutic alliance is a process that evolves across a course of treatment and a positive therapeutic alliance can be thought of as a necessary but not sufficient ingredient for effective treatment response. Indeed, across treatments for child mental health problems, therapeutic alliance has been found to be a predictor of treatment outcome (McLeod, 2011; Shirk, Karver, & Brown, 2011). A favorable therapeutic alliance may be needed to complete challenging therapy tasks, such as the exposure tasks that are critical for improvement. In recent years, research has emerged suggesting that there is a dynamic and reciprocal interplay across the course of CBT between therapeutic alliance and symptom improvement, such that positive therapeutic alliance predicts subsequently improved symptom response in treatment which, in turn, subsequently predicts further growth in therapeutic alliance and so on (e.g., Marker, Comer, Abramova, & Kendall, 2013).

Promoting a positive therapeutic alliance requires special consideration when working with youth populations. Importantly, youth typically do not self-refer to mental health services and many interventions are either advised or mandated by others. Children are often referred by parents, school, agency, court, or other social service provider for treatment of problems that they do not believe they have. It is not uncommon for children to have limited insight into their symptoms and, accordingly, they can be less motivated to collaborate in order to make improvements that others have deemed necessary.

In contrast to CBT for adults, supported CBTs for child problems often incorporate games, artwork, and other activities that children may enjoy. Importantly, in supported CBT the games and artwork are not the active ingredients of treatment. Whereas in play therapy and other

traditional forms of psychotherapy play can be seen as cathartic and therapeutic by itself, in supported protocols the play is included to promote positive engagement between the therapist and child (e.g., so the child looks forward to attending sessions and working with the therapist) and/or the CBT content is directly woven into these playful activities. For example, the CBT therapist might play a game of “feelings charades” with a child in order to build awareness in the context of a playful activity of emotions and their expressions.

Parents and other central adults in children’s lives are often involved in children’s treatment. The involvement of multiple therapy participants introduces further obstacles to therapeutic alliance in CBT for youth. Therapists must carefully attend to all of the treatment participants and their concerns. Children are very sensitive to whether the therapist is on their “side” or their parents’ “side,” and the therapist who does not carefully balance the needs and interests of both the child and the parents can inadvertently alienate one party or the other.

The optimal posture of the CBT therapist in child therapy merits special attention, as CBT therapists fulfill many different roles to promote treatment gains. Functionally, CBT therapists simultaneously serve as *teachers* giving knowledge and skills to their clients, as *health care providers* delivering expert services for improved functioning, and as *coaches* encouraging youth to challenge themselves with fun and creative activities. Kendall (2012b) specifies three different roles that CBT therapists should fulfill: the “collaborator/consultant,” the “diagnostician,” and the “educator.” The “collaborator/consultant” works with the child to set goals with the notion that the therapist is an expert on the CBT treatment techniques and the child is the expert on him or herself. The expertise of both the therapist and the child are equally needed to meet treatment goals. The “diagnostician” is continually observing, collecting, and combining information about the child and their current difficulties to fully understand the child’s symptoms and how best to progress in treatment. The therapist integrates information from multiple sources (e.g., from the child, his or her parents, and teachers) across the backdrop of development, family values and preferences, culture, socioeconomic status, and other family variables to best understand the child’s needs and plan treatment appropriately. The “educator” teaches children specific skills and tailors his or her teaching methods to the child’s individual needs.

As the effective CBT therapist integrates the roles of collaborator, diagnostician, and educator to guide children throughout the course of treatment, the term “coach” is often used to describe the optimal posture (or attitude) of the therapist. The metaphor of “therapist as coach” connotes a supportive yet exacting and challenging individual who can bring out the best in each child.

Therapy Participants and Settings

Children, particularly young children, are highly dependent on their environments and various adults in their environments (e.g., parents, school personnel). Accordingly, CBT for child problems often incorporates a broader range of participants and settings than encountered in typical CBT for adult problems. We now consider the roles of parents and school personnel in child CBT.

Parental Involvement

Parents and other caregivers are viewed as the primary agents of change in the lives of children and, as noted, young children may not have the cognitive capacity to understand and implement treatment techniques on their own. Consequently, parents often take a lead role in treatment. CBT therapists work with parents to reshape the primary context of their child's development by working to reduce maladaptive patterns of interaction (e.g., parental accommodation and reinforcement of anxious or disruptive behaviors).

Once treatment begins, parents may take different roles, such as consultant, collaborator, or coclient (Kendall, 2012b). Parents act as consultants by providing information for diagnoses, treatment goals, and progress. Parents act as collaborators by participating in the treatment process; they may be taught the skills the therapist uses in session to further implement them at home and guide their child in implementing the skills on their own. Lastly, parents can be participating in treatment as a coclient if the focus of treatment is also the parent's psychopathology (e.g., reducing parental anxiety or depression related to child symptoms). It is critical to establish clear expectations and orient parents to the necessary roles they may need to fulfill from the onset.

The optimal level of parental involvement in treatment varies across treatment protocols and the age of the child. For example, in CBT for older adolescents, parents might participate in as little as one or two parent-only sessions to collaborate on treatment goals. For younger children in CBT, the majority of the treatment may be parent training with or without child involvement.

School Personnel Involvement

Children spend the majority of their waking hours at school interacting with school personnel (e.g., teachers, counselors, paraprofessionals) and the majority of children who seek services initially seek them in schools (Burns et al., 1995). School personnel can be recruited for assessment purposes, as collaborators in the treatment process, or as direct treatment providers. Therapists commonly work with teachers to reinforce positive classroom behaviors (e.g., attending to the teacher,

finishing work, participating in class) and decrease negative or avoidant classroom behaviors (e.g., interrupting, aggression, social withdrawal). School personnel employ various techniques in order to aid in behavior change, including developing and monitoring reward charts and daily report cards (DRCs), carrying out exposure tasks in the school setting, and setting up classroom contingencies. Additionally, school personnel may employ more in-depth cognitive and social skills training. In a recent randomized trial by [Masia Warner et al. \(2016\)](#), school counselors implemented a group CBT program for youth with social anxiety with outcomes as favorable as those found for youth randomized to receive CBT from trained clinical psychologists.

Despite the added benefits of working within the school setting to promote and generalize behavior change in children, there are several obstacles to working within school settings. First, teachers and other school personnel are often overburdened with other tasks and adding additional responsibilities related to a child's mental health care may discourage them from participating in assessment and treatment. Moreover, a teacher's primary responsibility is meeting the educational needs of their students. As such, school-based CBT efforts should emphasize how targeting children's mental health will likely improve children's overall learning (e.g., treating a child's ADHD will prevent that child's disruptive symptoms from further disruptive the learning of other children in the classroom, treating a child's anxiety will help him or her demonstrate their full comprehension of classroom material on examinations).

CHILD AND ADOLESCENT CBT COMPONENTS FOR SPECIFIC PROBLEM AREAS

As we noted earlier, CBT protocols target associations between thoughts, behaviors, and emotions that are maintained by behavioral and/or cognitive processes and they typically entail some combination of eight key components of CBT practice: psychoeducation, routine symptom monitoring, somatic management, contingency management, behavioral engagement, cognitive restructuring and related thought challenging, problem-solving, and relapse prevention. That said, clinical trials have demonstrated that the optimal balance of these components, and the specific foci and structure of CBT, varies depending on the specific domain of child problems being targeted in treatment. To illustrate the heterogeneity and flexibility of modern CBT for various child problems, we now briefly present the most well-supported CBT practices for a number of the most common problem areas that bring youth into treatment: anxiety, depression, trauma, and attention-deficit/hyperactivity disorder and disruptive behavior problems. Although we provide an overview of these practices

here, more comprehensive details about specific protocols and session-by-session guides can be found elsewhere (e.g., [Kendall, 2012a](#)).

CBT for Child Anxiety

Although there are many overlapping and neighboring CBT protocols that target child anxiety, supported CBT protocols typically share the same core components: psychoeducation about child anxiety, relaxation strategies, cognitive restructuring, and exposure tasks.

CBT for child anxiety typically begins with psychoeducation about anxiety and teaches children (and parents if they are involved in treatment) that fear and anxiety are normal human responses to different stimuli in the environment. Children are taught that anxiety is normal and that it is only a problem when it becomes difficult to manage and interferes with everyday activities. Children are taught how their body physiologically responds to fear and that the feelings in their bodies, their maladaptive thoughts, and their behavior can all influence one another. Children are additionally taught that the more they confront and remain in age-appropriate anxiety-provoking situations, the more their ability to cope with anxiety will improve and they will feel more confident to face their fears in the future. Children are also taught relaxation strategies that target physiological responses to anxiety. Children often learn diaphragmatic breathing exercises, progressive muscle relaxation, and are engaged in guided imagery.

CBT for child anxiety typically then proceeds to focus on restructuring maladaptive cognitions. Children are taught to identify their dysfunctional thoughts in specific anxiety-provoking situations (e.g., a boy with social anxiety disorder might learn to recognize that when he is in the classroom he is commonly thinking to himself “if I answer the question wrong, everyone will laugh at me and think I’m dumb”). Anxious youth are then given skills to challenge the validity and accuracy of their thoughts and to reframe those thoughts to be more adaptive and accurate (e.g., for the socially anxious child a more adaptive thought might be “I’ve answered questions incorrectly before and nobody has laughed; I know the answer to the question and if I mess up, that’s okay—everybody makes mistakes sometimes!”). Problem-solving strategies are also emphasized in these earlier stages of CBT treatment for child anxiety.

Treatment then shifts to exposure tasks. Many experts suggest exposures to be the single most important element of effective CBT for child anxiety ([Kendall et al., 2005](#)) and, as such, exposure tasks typically dominate at least half of the treatment sessions in well-supported protocols. Exposure tasks have children gradually confront the situations and stimuli that they most fear and avoid. Typically, the child and therapist (and parents when involved in treatment) together create a

hierarchy of feared situations (e.g., a child with a specific phobia of dogs may put “seeing a dog on television or online” low on the hierarchy and may put “giving a dog a hug” at the top of the hierarchy). The CBT therapist encourages the child to use the coping strategies acquired in treatment in increasingly anxiety-provoking situations, always rewarding the child’s effort with social reinforcement (e.g., labeled praises) and/or tangible reinforcement (e.g., a sticker).

It has been noted that many providers who do not specialize in CBT regrettably shy away from exposure tasks when working with anxious youth, misperceiving that exposure tasks are inappropriate for youth and may interfere with treatment engagement and a positive therapeutic alliance. Importantly, [Kendall et al. \(2009\)](#) randomly assigned anxious youth to receive either psychoeducation/support or psychoeducation/support plus exposure tasks and found that the addition of exposure tasks did not rupture therapeutic alliance. In fact, growth curve modeling showed that therapeutic alliance for anxious youth who participated in exposure tasks actually continued to improve after the introduction of exposure tasks.

Some CBT protocols targeting child anxiety also involve parents directly in treatment, although for anxious youth 8 years and older it is not clear that including parents in treatment sessions actually improves outcomes ([Barmish & Kendall, 2005](#); [Breinholst, Esbjørn, Reinholdt-Dunne, & Stallard, 2012](#)). Importantly, however, every supported CBT protocol for anxiety in children ages 7 years and below takes a parent-focused approach ([Cartwright-Hatton et al., 2011](#); [Comer et al., 2012](#)). Given the evidence, it seems as though parental involvement in CBT is critical when treating anxiety in young children; for anxious youth in middle childhood or adolescence, parental involvement is not always critical but can be helpful when parents display maladaptive patterns of symptom accommodation and overprotection.

When working with younger anxious children (i.e., ages 7 years and below), CBT typically de-emphasizes cognitive strategies to accommodate the more restricted cognitive capacities of early childhood. CBTs targeting anxiety in early childhood utilize parents as the agents of change. Programs such as the Coaching Approach Behavior and Leading by Modeling Program ([Comer et al., 2012](#); [Puliafico, Comer, & Albano, 2013](#)) teach parents how to reduce symptom accommodation, overprotection, and other child anxiety-maintaining behaviors, and directly coach parents to engage their young child in exposure tasks.

CBT for Child Depression

CBT for depressive disorders in children and adolescents incorporates many of the same components of CBT for anxiety and, as with

CBT for anxiety, there are many overlapping and neighboring CBT protocols that target youth depression that have shown empirical support. Many CBT protocols are family-based, incorporating parents, and sometimes other members of the family, into treatment. Including parents in treatment can be particularly helpful when youth depression is triggered by stressful life/family events.

CBT for youth depression typically incorporates some combination of psychoeducation, problem formulation and goal setting, mood monitoring, behavioral activation/activity scheduling, relaxation strategies, social skills building, problem solving, and cognitive restructuring (Reinecke & Ginsburg, 2008). Psychoeducation entails giving structured information to the child or adolescent (and their parent if involved in treatment) about biological, cognitive, and environmental factors that can be associated with depression, how various factors interact to influence symptoms, and what to expect in a course of CBT for youth depression. Within the first few sessions, the CBT therapist engages the child in problem formulation and goal setting—the therapist and child collaboratively generate a list of problems that the youth is currently experiencing that seem linked with depressive symptoms. *Mood monitoring* is another critical early component of CBT for youth depression. Youth are taught to engage in “detective thinking” to identify their emotions and track their feelings, both positive *and* negative. Youth are taught to identify specific thoughts that accompany their feelings.

Behavioral activation and activity scheduling can be an essential component of CBT for youth depression, particularly for more withdrawn depressed youth or youth under stress. Depressed children or adolescents are encouraged to schedule and participate in activities that are pleasurable and social to improve their mood. The youth and therapist generate a list of realistic activities in which the youth can engage every day; it is expected that increasing participation in pleasurable activities will naturally increase environmental reinforcements.

As in CBT for child anxiety, youth in CBT for depression are often taught relaxation strategies such as diaphragmatic breathing and progressive muscle relaxation. Youth are encouraged to use relaxation strategies during times of mood dysregulation and stressful situations. Focusing on relaxation and affect regulation can help direct youth away from maladaptive responses such as rumination and poor thinking patterns. Depressive symptoms commonly have a negative impact on interpersonal relationships and, accordingly, social skills training can be another commonly used CBT component for youth depression. Social skills building for depressed youth teaches children and adolescents how to more appropriately interact with others via role plays and

guided practice. For example, youth practice appropriate body language and conversation skills and are taught to reduce excessive reassurance-seeking in conversation.

Problem-solving components of CBT for youth depression provides children and adolescents with strategies to resolve specific difficulties or dilemmas in their interpersonal lives via role-plays and in-session practice. The RIBEYE acronym is frequently used to facilitate recall of problem-solving strategies: Relax, Identify the problem, Brainstorm possible solutions, Evaluate their strengths and weaknesses, say Yes to one (or two), and Encourage oneself for success (Reinecke & Ginsburg, 2008).

Finally, cognitive components of CBT for youth depression often target negative views the child or adolescent holds of themselves, others/the world, and their future. Cognitive restructuring aims to reframe maladaptive perspectives into more positive and adaptive thinking. Youth are encouraged to identify automatic thoughts and cognitive distortions (e.g., catastrophizing, jumping to conclusions) and challenge the validity of distorted thoughts. Subsequently, youth are encouraged to generate counter-thoughts to more adaptively interpret and think about stressful situations.

CBT for Child Trauma

CBT for youth who have experienced trauma, or trauma-focused CBT (TF-CBT; Cohen, Mannarino, & Deblinger, 2012; Cohen, Mannarino, Perel, & Staron, 2007; Scheeringa, Weems, Cohen, Amaya-Jackson, & Guthrie, 2011), emphasizes making meaning of traumatic experiences and coping with the reminders of and stressors related to traumatic experiences. Family involvement can be particularly important in TF-CBT, especially when family members are also deeply impacted by the traumas their children have experienced. Parents are taught skills, such as active listening, and strategies to make the child feel safe in their environment. TF-CBT emphasizes a strong therapeutic relationship with the assumption that these youth and families may have a hard time discussing their traumatic experiences and must feel they are in a safe and accepting environment in order to make use of treatment.

Psychoeducation in TF-CBT focuses on normalizing responses to trauma as well as information on the trauma itself; for example, families might be provided with information on the prevalence and impact of a specific trauma endured. Psychoeducation about trauma and youth reactions can also help begin to make the child feel more comfortable discussing the traumatic experience and being in treatment. TF-CBT therapists also incorporate relaxation strategies, such as diaphragmatic breathing, progressive muscle relaxation, and guided

imagery to help youth calm their bodies during stressful times or when recalling or remembering traumatic events. Youth often experience a confusing array of emotions in response to trauma. Accordingly, affective expression is a key component of TF-CBT that helps youth identify emotions and connect situations to certain emotions those situations might evoke. In addition, youth are encouraged to use specific coping strategies to help manage their emotions (e.g., listening to music, going on a walk).

As treatment shifts to focus on the specific traumatic experiences, developing a *trauma narrative* becomes a central component of TF-CBT. Youth are asked to generate a narrative discussing the specific occurrences of the trauma and any relevant feelings or thoughts experienced during the trauma. Youth are encouraged to gradually and repeatedly expose themselves to their traumatic memories until they become increasingly desensitized to those memories, and the memories become less challenging to tolerate. Exposure is also used to target avoidant behaviors related to their traumatic experience. Youth are encouraged to practice going to certain places or doing specific things that they have come to avoid due to painful reminders of their traumatic experience. As in CBT for other child problems, cognitive restructuring in TF-CBT helps youth identify and adjust maladaptive thinking patterns. Youth are encouraged to identify faulty and inaccurate thoughts related to their trauma and memories of the trauma and replace them with more positive and valid thoughts.

CBT for Disruptive Behavior Problems and Attention-Deficit/Hyperactivity Disorder

Most supported CBT protocols for disruptive behavior problems in children and ADHD place a predominant focus on the behavioral aspects of treatment and work to change the environments serving to maintain problem behaviors. For example, many target parenting behavior and family interaction patterns, effective commands and adaptive discipline techniques, and/or contingency management in the classroom (Eyberg, Nelson, & Boggs, 2008; Pelham & Fabiano, 2008). Changing the child's environment is emphasized over changing the child's cognitions because children with disruptive behavior problems and ADHD often have poor insight into their symptoms and how they may be impairing their interpersonal and academic lives. Their attention problems may also interfere with their ability to directly engage in cognitive therapy.

CBTs that work to disrupt coercive cycles of negative reinforcement in family interactions that promote oppositional and aggressive behaviors are collectively referred to as "behavioral parent training"

or “parent management training.” These programs focus on giving parents the skills they need to better manage their child’s behavior. Most protocols place importance on having parents give positive attention, such as praise, to their child for appropriate behaviors in order to build a positive relationship with their child. Prior to treatment, many children with ADHD and disruptive behavior problems are receiving extensive attention from the adults and peers in their lives for negative behaviors and positive behaviors are going unnoticed. Behavioral parent training teaches parents to use differential attention to child behaviors, such that they are giving positive attention (e.g., labeled praise) to positive behaviors and ignoring instances of minor misbehavior. Behavioral parent training for child disruptive behavior problems also gives parents disciplinary strategies and guides parents in how to give children effective direct commands and follow through with praise for compliance or a consistent time out procedure for noncompliance.

A DRC may be used in school settings to track child behavior and reinforce appropriate behavior through a point system. The DRC also serves as a way for parents and teachers to regularly communicate with one another about the child’s behavior. A small number of specific areas are selected for improvement (e.g., child gets aggressive) and three to eight related target behaviors are identified (e.g., plays without fighting at recess). Target behaviors are evaluated at regular intervals throughout the day and a home-based reward system is set such that parents provide daily and weekly rewards as appropriate for the child meeting specified target goals in school.

CBTs for older children and adolescents with disruptive behavior problems or ADHD typically engage children more directly and incorporate more cognitive components. One well-supported program for disruptive children between 3rd and 7th grade is the Coping Power program (Lochman, Wells, & Lenhart, 2008), which targets aggression and can be implemented in a group or individual format. In addition to providing parent training groups, Coping Power provides child group sessions (typically offered in the school setting) that teach youth skills in anger management, social problem-solving, and resisting peer-pressure. For adolescents with ADHD, there is growing support for the Supporting Teens’ Autonomy Daily Program (STAND; Sibley et al., 2013), which focuses on academics, organization, and time management, and encourages parents and adolescents to work together to fix problematic behaviors, monitor progress, and reinforce good behavior and skill usage. STAND also encourages parents and teachers to communicate to monitor the adolescent’s academic work and behavior.

RECENT ADVANCES AND INNOVATIONS

Despite the enormous success of CBT practices for treating child and adolescent problems in specialty care settings and research laboratories, our current mental health service delivery models fail to reach adequate numbers of affected youth. With hundreds of controlled evaluations having now demonstrated overwhelming support for the efficacy of CBT for a broad range children's mental health problems (Kendall, 2012a; Weisz, Weiss, Han, Granger, & Morton, 1995), the next great challenge is to develop innovative solutions for meaningfully broadening the reach of quality CBT for youth in need. Indeed, systematic barriers prevent large numbers of children from receiving timely care and gaps persist between CBT provided in specialty care settings and various services provided in the settings where the majority of youth actually receive mental health care (Comer & Barlow, 2014; Sandler, Ostrom, Bitner, Ayers, & Wolchik, 2005; Weisz, Sandler, Durlak, & Anton, 2005).

For most affected youth, problems persist in the availability, accessibility, and acceptability of quality CBT. Regarding CBT availability, there are inadequate numbers of CBT-trained therapists providing care in frontline practice settings. Professional workforce shortages in mental health care abound, with a considerable proportion of U.S. counties lacking any child psychologist, psychiatrist, or social worker (Comer, 2015). Problems in CBT availability for youth are particularly problematic in remote regions, with over three-quarters of federally designated Mental Health Professional Shortage Areas situated in rural areas (Bird, Dempsey, & Hartley, 2001; Comer, Elkins, Chan, & Jones, 2014; National Advisory Committee on Rural Health, 2002). Long wait lists at poorly funded clinics considerably slow the speed of service delivery. Comer and Barlow (2014) have suggested that collectively these factors may help to explain, in part, recent national trends showing that psychotherapies such as CBT have assumed a decreasingly prominent role in outpatient care (Olfson & Marcus, 2010). Poor quality of care in frontline practice settings presents a further obstacle to CBT availability for many children and adolescents, as those receiving mental health care are not necessarily receiving supported services such as CBT. Treatments receiving the strongest support, such as CBT, are rarely disseminated effectively on a broad level.

We now turn our attention to four recent advances in the field that are working to transform the availability of quality CBT for affected children and adolescents. These innovations include transdiagnostic, modular, and intensive CBT formats, as well as the promising role of technology for improving the accessibility of CBT for traditionally underserved child populations.

Transdiagnostic CBT

The development of traditional CBT manuals over the past two decades has yielded a proliferation of “single-disorder protocols” that provide guided intervention for single mental health diagnoses and their clinical correlates. It has been noted that many of these single-disorder protocols have only minor and somewhat trivial variations from one another. Although the development of these manuals was intended, in part, to support CBT training, single-disorder protocols have grown so numerous that it has now become unnecessarily burdensome and near impossible for practicing clinicians to become familiar with, let alone to develop adequate proficiency in, every supported CBT manual. Moreover, it has not been clear how best to choose among or prioritize these single-disorder protocols for training. This state of affairs has regrettably hampered efforts at widespread dissemination and uptake of CBT (Barlow, Sauer-Zavala, Carl, Bullis, & Ellard, 2014).

In contrast to single-disorder protocols, transdiagnostic CBTs for youth focus on similarities and overlying features across child disorders, particularly those from neighboring classes of diagnoses and showing high levels of comorbidity. Although the early roots of behavioral and cognitive therapies, as well as client-centered and psychoanalytic approaches, were intrinsically transdiagnostic (see Comer, Elkins et al., 2014), it is now in light of major advances in knowledge in the affective neuroscience, underlying etiology, and latent structure of disorders (see Barlow, 2004; Wilamowska et al., 2010) that research groups have been able to develop unifying CBT protocols that can be flexibly applied across a range of diagnoses sharing common components and etiological mechanisms. For example, transdiagnostic CBT protocols simultaneously targeting the range of anxiety and unipolar mood disorders in youth are beginning to show strong promise (see Chu, 2012; Ehrenreich, Goldstein, Wright, & Barlow, 2009; Weersing, Rozenman, Maher-Bridge, & Campo, 2012). A downward extension of the unified protocol (UP) for transdiagnostic treatment of emotional disorders (Barlow et al., 2010) is an emotion-focused CBT for application to all internalizing child disorders that was developed by distilling and incorporating the common principles among existing child CBTs—namely restructuring maladaptive cognitions, changing maladaptive action tendencies, preventing emotional avoidance, and engaging emotion exposure procedures (Ehrenreich-May et al., 2008). In addition, the UP for youth places great emphasis on the adaptive, functional nature of emotions, building the child’s awareness of physical sensations, cognitions, and behaviors and their relationships with emotional experiences and ability to identify and adjust maladaptive reactions to these experiences.

Transdiagnostic CBT approaches for treating child mental health problems have begun to show promising empirical support in a growing set of clinical evaluations (Bilek & Ehrenreich-May, 2012; Ewing, Monsen, Thompson, Cartwright-Hatton, & Field, 2015). In the event of continued evidence of success for transdiagnostic CBTs, among the many advantages includes a more parsimonious application of CBT and a dramatic reduction in the number of CBT protocols in which we would expect providers to be well versed. These multidisorder interventions consolidate traditional CBT components across diagnostic boundaries for flexible implementation, allowing for more resource-efficient dissemination and implementation with the added advantage of simultaneously addressing comorbid problems. Transdiagnostic CBTs may indeed have the potential to make dissemination efforts less burdensome, thus elevating the public health significance of the availability of supported CBTs (McHugh & Barlow, 2010).

Modular CBT

Another increasingly popular approach to combatting the single-disorder manual proliferation barrier to broad dissemination of children's CBT has been modular approaches to CBT delivery. Whereas traditional CBT protocols apply treatment components in a bound and linear format, it has become apparent that not every child will benefit comparably from each and every treatment component in an indicated CBT (Chorpita, Daleiden, & Weisz, 2005). And, of course, CBT providers encounter a broad range of sometimes unrelated, comorbid problems across internalizing and externalizing dimensions. Thus, many children may benefit from various treatment elements found across a range of single-diagnosis CBT protocols. For the heterogeneous cases that populate practice settings, employing a linear sequence of single-diagnosis CBT manuals may not be optimal from either a dissemination or treatment efficiency standpoint. Modular treatments address such problems through strategic treatment redesign—procedures from supported CBTs for specific identified problems are structured as free-standing modules and decision flowcharts based on individual client characteristics guide module selection and individualized treatment component sequencing (Chiu et al., 2013; Chorpita & Weisz, 2005; Weisz et al., 2012).

Modular approaches to child CBT delivery are already showing success in controlled evaluations, with modular CBT and dynamic treatment regimens (i.e., treatments that systematically adapt to children's shifting symptom presentations) distinguishing themselves from usual care in practice settings (Weisz et al., 2012). For example, the MATCH (modular approach to therapy for children) program produced steeper

improvement trajectories in a broad clinical child sample than usual care and standard linear single-disorder CBT protocols. Following treatment, MATCH was also linked with significantly fewer children meeting criteria for their presenting diagnosis than usual care, whereas outcomes of standard single-disorder CBTs did not differ from usual care outcomes (Weisz et al., 2012). Importantly, therapists trained in standard linear single-disorder CBTs held more negative views of treatment manuals than did therapists trained in supported modular programs that incorporate potentially shifting treatment needs and tailor treatment sequences to individual child presentations (Borntrager, Chorpita, Higa-McMillan, & Weisz, 2009).

Intensive CBT

Given regional shortages in the availability of quality CBT (see Comer, Elkins et al., 2014; Elkins, McHugh, Santucci, & Barlow, 2011), it has become increasingly common for CBT specialty providers to offer intensive treatment formats in their practices. This typically requires that families travel for brief (e.g., 1–3 week) periods of all-day sessions of CBT not offered in their local community. Intensive treatments can deliver an entire course of a CBT in a shorter period of time through longer individual sessions (Comer, Elkins et al., 2014; Ehrenreich & Santucci, 2009) and, as such, can be preferable over traditional weekly CBT with respect to the necessary time commitment, travel requirements, and reduction of stigma. Intensive treatments may be conducted during times convenient for the family, such as summer or holiday breaks, during which times children have fewer academic demands. Group intensive CBT formats, in particular, have grown in popularity for children as they can help children feel less isolated by their condition in the context of peers experiencing similar symptoms. Peer interaction may also provide unique opportunities for modeling and practice. Summer “camp-based” intensive group CBTs, in particular, have risen in popularity (e.g., Carpenter et al., 2014; Santucci & Ehrenreich-May, 2013). To date, clinical trials evaluating intensive CBTs have shown positive support in the treatment of child anxiety disorders (e.g., Gallo, Cooper-Vince, Hardaway, Pincus, & Comer, 2014; Hardway, Pincus, Gallo, & Comer, 2015; Ollendick et al., 2009), OCD (e.g., Storch et al., 2007), ADHD (e.g., Sibley, Smith, Evans, Pelham, & Gnagy, 2012), and selective mutism (Carpenter et al., 2014).

Technology and CBT

Recent advances in technology and the rapid uptake of both advanced consumer products and Internet use have yielded a potentially

transformative opportunity to reconsider the means by which supported clinic-based CBTs are provided. Given the far-reaching functionality of personal and home-based technologies for transmitting and collecting rich information across a variety of formats, current research efforts have worked to leverage technology in a variety of ways to develop innovative means by which to expand the accessibility to quality CBT (see [Comer, 2015](#); [Myers & Comer, 2016](#)). In fact, according to Web of Science, there were 145 scientific publications between 2000 and 2014 that addressed “child telemental health” (and/or “child behavioral telehealth,” “child telepsychology,” or “child telepsychiatry”) and roughly half of these publications were printed in just the past 5 years ([Comer & Myers, 2016](#)).

Various platforms have been leveraged to expand the scope and reach of CBT—including PCs, smartphones, and tablets—and technology-based programs range from those that are fully self-administered to those incorporating minimal therapist involvement to those entailing full therapist involvement. Stand-alone (no therapist involvement) computer-based programs and software create platforms through which CBT concepts can be more widely disseminated and self-implemented ([Kendall, Khanna, Edson, Cummings, & Harris, 2011](#); [Khanna & Kendall, 2008](#)). Increasing research is finding that client continued use and adherence to self-administered technology-based CBTs can be problematic in the absence of some degree of involvement and check-ins from a professional or paraprofessional ([Mohr, Cuijpers, & Lehman, 2011](#)). What the nature of that involvement optimally entails is still unclear, but research is pointing to the importance of “supportive accountability” that has clients’ sustained participation in and adherence to technology-based programs is improved when clients feel accountable to a coach who is seen as trustworthy, benevolent, and having expertise ([Mohr et al., 2011](#)). Such findings have led to increased incorporation of minimal therapist involvement formats of technology-based services, in which a supportive coach may briefly check in on a semi-regular basis (e.g., weekly or bi-weekly phone calls or emails) to discuss the content of the program, assess adherence to assignments, and answer any questions the child or family may have. Moreover, smartphones and mobile apps have been increasingly used as adjunctive components to clinic-based CBT, collecting ecologically valid data about children’s and adolescents’ symptoms and impairments *in situ* and affording asynchronous communication between therapists and clients ([Pramana, Pramanto, Kendall, & Silk, 2014](#); [Whiteside, 2016](#)).

The use of videoconferencing to deliver real-time CBT has been one of the fastest growing trends in CBT practice and research ([Comer, 2015](#); [Crum & Comer, 2016](#)). For mental health care, a discipline that relies primarily on verbal communication and visual observation, videoconferencing methods can offer strategic opportunities to overcome geographical barriers to quality CBT by extending the availability of CBT providers.

Children dwelling in rural or impoverished regions—areas typically beset by insufficient availability of CBT care—can participate in real-time interactive CBT conducted by experts, regardless of geographic proximity to a CBT clinic. A growing body of empirical work supports the preliminary efficacy, feasibility, tolerability, sustainability, and efficacy of applying videoconferencing methods to remotely deliver CBT to children and families in need (Comer et al., 2015; Comer, Furr et al., 2014; Crum & Comer, 2016; Nelson, Barnard, & Cain, 2003).

FUTURE DIRECTIONS AND CONCLUDING THOUGHTS

Children's mental health problems are highly prevalent and impose enormous burdens at the individual, family, and societal level. Fortunately, the past few decades have witnessed tremendous advances in the development and evaluation of CBT procedures with demonstrated success in treating a considerable share of children's mental health problems. Simple downward extensions of treatments designed for older individuals to address the clinical needs of youth are misguided. Developmentally informed models of care that account for the cognitive abilities of youth and that incorporate special considerations inherent in working with child populations are increasingly showing great success. In recent years, advances in transdiagnostic and modularized CBT for youth have been working to improve the efficiency, acceptability, and uptake of CBT for children and adolescents. Researchers and providers are also increasingly leveraging new technologies to meaningfully expand the scope and reach of CBT for youth.

Given these extraordinary advances in developing developmentally sensitive CBT procedures for youth, the challenge ahead will now be to improve training models and dissemination procedures in order to maximize the quality and availability of CBT for the majority of youth in need. Considerable attention and large financial commitments of over several billion dollars have focused in recent years on innovative solutions to the problems of CBT availability and quality (McHugh & Barlow, 2010). Formal evaluations of large-scale dissemination and implementation programs are ongoing, but early efforts do reveal some consistent findings—quality dissemination efforts are expensive and hard to sustain. The larger, more successful programs that have incorporated didactics and extended competency training and consultation have cost hundreds of millions of dollars each and many still have not actualized the large and lasting impacts aspired to at initiative outset (Comer & Barlow, 2014). Regrettably, the present availability of quality CBT for youth is largely concentrated in major academic hubs and

metropolitan regions. Accordingly, future research that supports cost-efficient models for promoting effective and sustainable CBT training on a broad scale is critical to optimize the public health impact of the past several decades of successful research on CBT for children's mental health problems.

References

- Abikoff, H., Gallagher, R., Wells, K. C., Murray, D. W., Huang, L., Lu, F., & Petkova, E. (2013). Remediating organizational functioning in children with ADHD: Immediate and long-term effects from a randomized controlled trial. *Journal of Consulting and Clinical Psychology, 81*, 113–128.
- Barlow, D. H. (2004). Psychological treatments. *American Psychologist, 59*, 869–878.
- Barlow, D. H., Farchione, T. J., Fairholme, C. P., Ellard, K. K., Boisseau, C. L., Allen, L. B., & Ehrenreich-May, J. T. (2010). *Unified protocol for transdiagnostic treatment of emotional disorders: Therapist guide*. New York, NY: Oxford University Press.
- Barlow, D. H., Sauer-Zavala, S., Carl, J. R., Bullis, J. R., & Ellard, K. K. (2014). The nature, diagnosis, and treatment of neuroticism: Back to the future. *Clinical Psychological Science, 2*, 344–365. Available from <http://dx.doi.org/10.1177/2167702613505532>.
- Barmish, A. J., & Kendall, P. C. (2005). Should parents be co-clients in cognitive-behavioral therapy for anxious youth? *Journal of Clinical Child and Adolescent Psychology, 34*, 569–581.
- Beck, A. T. (1976). *Cognitive therapy and the emotional disorders*. New York: New American Library.
- Bilek, E., & Ehrenreich-May, J. (2012). An open trial investigation of a transdiagnostic treatment for children with anxiety and depressive symptoms. *Behavior Therapy, 43*, 887–897.
- Bird, D. C., Dempsey, P., & Hartley, D. (2001). *Addressing mental health workforce needs in underserved rural areas: Accomplishments and challenges*. Portland, ME: Maine Rural Health Research Center.
- Borntrager, C. F., Chorpita, B. F., Higa-McMillan, C., & Weisz, J. R. (2009). Provider attitudes toward evidence-based practices: Are the concerns with the evidence or with the manuals? *Psychiatric Services, 60*, 677–681. doi: 10.1176/appi.ps.60.5.677.
- Boswell, J. F., Sharpless, B. A., Greenberg, L. S., Heatherington, L., Huppert, J. D., Barber, J. P., ... Castonguay, L. G. (2011). Schools of psychotherapy and the beginning of a scientific approach. In D. H. Barlow (Ed.), *The Oxford handbook of clinical psychology*. New York: Oxford University Press.
- Breinholst, S., Esbjørn, B. H., Reinholdt-Dunne, M. L., & Stallard, P. (2012). CBT for the treatment of child anxiety disorders: A review of why parental involvement has not enhanced outcomes. *Journal of Anxiety Disorders, 26*, 416–424.
- Burns, B. J., Costello, E. J., Angold, A., Tweed, D., Stangl, D., Farmer, E. M. Z., & Erkanli, A. (1995). Children's mental health service use across service sectors. *Health Affairs, 14*, 147–159.
- Carpenter, A. L., Puliafico, A. C., Kurtz, S. M. S., Pincus, D. B., & Comer, J. S. (2014). Extending parent-child interaction therapy for early childhood internalizing problems: New advances for an overlooked population. *Clinical Child and Family Psychology Review, 17*, 340–356.
- Cartwright-Hatton, S., McNally, D., Field, A. P., Rust, S., Laskey, B., Dixon, C., ... Woodham, A. (2011). A new parenting-based group intervention for young anxious children: Results of a randomized controlled trial. *Journal of the American Academy of*

- Child and Adolescent Psychiatry*, 50, 242–251. Available from <http://dx.doi.org/10.1016/j.jaac.2010.12.015>.
- Chiu, A. W., Langer, D. A., McLeod, B. D., Har, K., Drahota, A., Galla, B. M., ... Wood, J. J. (2013). Effectiveness of modular CBT for child anxiety in elementary schools. *School Psychology Quarterly*, 28, 141–153. Available from <http://dx.doi.org/10.1037/spq0000017>.
- Chorpita, B. F., Daleiden, E. L., & Weisz, J. R. (2005). Modularity in the design and application of therapeutic interventions. *Applied and Preventive Psychology*, 11, 141–156.
- Chorpita, B. F., & Weisz, J. R. (2005). *Modular approach to therapy for children with anxiety, depression, or conduct problems*. Honolulu, HI: University of Hawaii at Manoa.
- Chu, B. (2012). Translating transdiagnostic approaches to children and adolescents. *Cognitive and Behavioral Practice*, 19, 1–4.
- Cohen, J. A., Mannarino, A. P., & Deblinger, E. (Eds.) (2012). *Trauma-focused CBT for children and adolescents: Treatment applications*. New York, NY: Guilford Press.
- Cohen, J. A., Mannarino, A. P., Perel, J. M., & Staron, V. (2007). A pilot randomized controlled trial of combined trauma-focused CBT and sertraline for childhood PTSD symptoms. *Journal of the American Academy of Child & Adolescent Psychiatry*, 46, 811–819.
- Comer, J. S. (2015). Introduction to the special section: Applying new technologies to extend the scope and accessibility of mental health care. *Cognitive and Behavioral Practice*, 22, 253–257.
- Comer, J. S., & Barlow, D. H. (2014). The occasional case against broad dissemination and implementation: Retaining a role for specialty care in the delivery of psychological treatments. *American Psychologist*, 69, 1–18.
- Comer, J. S., Blanco, C., Grant, B., Hasin, D., Liu, S. M., Turner, J. B., & Olfson, M. (2011). Health-related quality of life across the anxiety disorders: Results from the National Epidemiologic Survey on Alcohol and Related Conditions. *Journal of Clinical Psychiatry*, 72, 43–50.
- Comer, J. S., Elkins, R. M., Chan, P. T., & Jones, D. J. (2014). New methods of service delivery for children's mental health care. In C. A. Alfano, & D. Beidel (Eds.), *Comprehensive evidence-based interventions for school-aged children and adolescents*. New York: Wiley.
- Comer, J. S., Furr, J. M., Cooper-Vince, C., Kerns, C. E., Chan, P. T., Edson, A. L., ... Freeman, J. B. (2014). Internet-delivered, family-based treatment for early-onset OCD: A preliminary case series. *Journal of Clinical Child and Adolescent Psychology*, 43(1), 74–87. Available from <http://dx.doi.org/10.1080/15374416.2013.855127>.
- Comer, J. S., Furr, J. M., Cooper-Vince, C., Madigan, R. J., Chow, C., Chan, P. T., ... Eyberg, S. M. (2015). Rationale and considerations for the Internet-based delivery of parent-child interaction therapy. *Cognitive and Behavioral Practice*, 22, 302–316.
- Comer, J. S., & Myers, K. M. (2016). Future directions in the use of telemental health to improve the accessibility and quality of children's mental health services. *Journal of Child and Adolescent Psychopharmacology*, 26, 296–300.
- Comer, J. S., Puliafico, A. C., Aschenbrand, S. G., McKnight, K., Robin, J. A., Goldfine, M. E., & Albano, A. M. (2012). A pilot feasibility evaluation of the CALM program for anxiety disorders in early childhood. *Journal of Anxiety Disorders*, 26, 40–49. Available from <http://dx.doi.org/10.1016/j.janxdis.2011.08.011>.
- Costello, E. J., Egger, H., & Angold, A. (2005). 10-year research update review: The epidemiology of child and adolescent psychiatric disorders, I: Methods and public health burden. *Journal of the American Academy of Child and Adolescent Psychiatry*, 44, 972–986.
- Crum, K. I., & Comer, J. S. (2016). Using synchronous videoconferencing to deliver family-based mental health care. *Journal of Child and Adolescent Psychopharmacology*, 26, 229–234.
- Devine, R. T., & Hughes, C. (2013). Silent films and strange stories: Theory of mind, gender, and social experiences in middle childhood. *Child Development*, 84, 989–1003.

- DiGiuseppe, R., Linscott, J., & Jilton, R. (1996). Developing the therapeutic alliance in child-adolescent psychotherapy. *Applied and Preventive Psychology, 5*, 85–100.
- Ehrenreich, J. T., Goldstein, C. R., Wright, L. R., & Barlow, D. H. (2009). Development of a unified protocol for the treatment of emotional disorders in youth. *Child and Family Behavior Therapy, 31*, 20–37. doi: 10.1080/07317100802701228.
- Ehrenreich, J. T., & Santucci, L. C. (2009). Special series: Intensive cognitive-behavioral treatments for child and adolescent anxiety disorders. *Cognitive and Behavioral Practice, 16*, 290–293. doi: 10.1016/j.cbpra.2009.04.001.
- Ehrenreich-May, J., Buzzella, B.A., Trosper, S.E., Bennett, S.M., Wright, L.A. & Barlow, D. H. (2008). *Unified protocol for the treatment of emotional disorders in youth*. Unpublished treatment manual, University of Miami and Boston University.
- Elkins, R. M., McHugh, R. K., Santucci, L. C., & Barlow, D. H. (2011). Improving the transportability of CBT for internalizing disorders in children. *Clinical Child and Family Psychology Review, 14*, 161–173. Available from <http://dx.doi.org/10.1007/s10567-011-0085-4>.
- Ellis, A. (1962). *Reason and emotion in psychotherapy*. New York: Lyle Stuart.
- Ewing, D. L., Monsen, J. J., Thompson, E. J., Cartwright-Hatton, S., & Field, A. (2015). A meta-analysis of transdiagnostic cognitive behavioural therapy in the treatment of child and young person anxiety disorders. *Behavioural and Cognitive Psychotherapy, 43*, 562–577. Available from <http://dx.doi.org/10.1017/S1352465813001094>.
- Eyberg, S. M., Nelson, M. M., & Boggs, S. R. (2008). Evidence-based psychosocial treatments for children and adolescents with disruptive behavior. *Journal of Clinical Child and Adolescent Psychology, 37*, 215–237.
- Flavell, J. H., Green, F. L., & Flavell, E. R. (1998). The mind has a mind of its own: Developing knowledge about mental uncontrollability. *Cognitive Development, 13*, 127–138.
- Flavell, J. H., Green, F. L., & Flavell, E. R. (2000). Development of children's awareness of their own thoughts. *Journal of Cognition and Development, 1*, 97–112.
- Flavell, J. H., Miller, P. H., & Miller, S. A. (2001). *Cognitive development* (4th Ed.). New York, NY: Prentice Hall.
- Gallo, K. P., Cooper-Vince, C. E., Hardway, C., Pincus, D. B., & Comer, J. S. (2014). Trajectories of change across outcomes in intensive treatment for adolescent panic disorder and agoraphobia. *Journal of Clinical Child and Adolescent Psychology, 43*, 742–750.
- Grave, J., & Blissett, J. (2004). Is cognitive behavior therapy developmentally appropriate for young children? A critical review of the evidence. *Clinical Psychology Review, 24*, 399–420. Available from <http://dx.doi.org/10.1016/j.cpr.2004.03.002>.
- Grazzani, I., & Ornaghoi, V. (2012). How do use and comprehension of mental-state language relate to theory of mind in middle childhood? *Cognitive Development, 27*, 99–111.
- Hardway, C. L., Pincus, D. B., Gallo, K. P., & Comer, J. S. (2015). Parental involvement in intensive treatment for adolescent panic disorder and its impact on depression. *Journal of Child and Family Studies, 24*, 3306–3317.
- Hofmann, S. G. (2011). *An introduction to modern CBT: Psychological solutions to mental health problems*. New York: Wiley-Blackwell.
- Hofmann, S. G., Doan, S. N., Sprung, M., Wilson, A., Ebessutani, C., Andrews, L. A., ... Harris, P. L. (2016). Training children's theory-of-mind: A meta-analysis of controlled studies. *Cognition, 150*, 200–212.
- Hofmann, S. G., Sawyer, A. T., Fang, A., & Asnaani, A. (2012). Emotion dysregulation model of mood and anxiety disorders. *Depression and Anxiety, 29*, 409–416.
- Kazantzis, N., Deane, F. P., & Ronan, K. R. (2000). Homework assignments in cognitive and behavioral therapy: A meta-analysis. *Clinical Psychology: Science and Practice, 7*, 189–202.
- Kendall, P. C. (2012a). *Child and adolescent therapy: Cognitive-behavioral procedures* (4th Ed.). New York: Guilford Press.

- Kendall, P. C., Comer, J. S., Marker, C. D., Creed, T. A., Puliafico, A. C., Hughes, A. A., . . . Hudson, J. L. (2009). In-session exposure tasks and therapeutic alliance across the treatment of childhood anxiety disorders. *Journal of Consulting and Clinical Psychology*, 77, 517–525. Available from <http://dx.doi.org/10.1097/CHI.0b013e31817eed2f>.
- Kendall, P. C., Khanna, M. S., Edson, A., Cummings, C., & Harris, M. S. (2011). Computers and psychosocial treatment for child anxiety: Recent advances and ongoing efforts. *Depression and Anxiety*, 28, 58–66.
- Kendall, P. C. (2012b). Guiding theory for therapy with children and adolescents. In P. C. Kendall (Ed.), *Child and adolescent therapy: Cognitive-behavioral procedures* (4th Ed.). New York: Guilford Press.
- Kendall, P. C., Robin, J. A., Bertzos, K., Suveg, C., Flannery-Schroeder, E., & Gosch, E. (2005). Considering CBT with anxious youth? Think exposures. *Cognitive and Behavioral Practice*, 12, 136–148.
- Kessler, R. C., Avenevoli, S., Costello, E. J., Georgiades, K., Green, J. G., Gruber, M. J., . . . Merikangas, K. R. (2012). Prevalence, persistence, and sociodemographic correlates of DSM-IV disorders in the National Comorbidity Survey Replication Adolescent Supplement. *Archives of General Psychiatry*, 69, 372–380.
- Kessler, R. C., Berglund, P., Demler, O., Jin, R., Merikangas, K. R., & Walters, E. E. (2005). Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the National Comorbidity Survey Replication. *Archives of General Psychiatry*, 62, 593–602.
- Khanna, M. S., & Kendall, P. C. (2008). Computer-assisted CBT for child anxiety: The coping cat CD-ROM. *Cognitive and Behavioral Practice*, 15, 159–165. Available from <http://dx.doi.org/10.1016/j.cbpra.2008.02.002>.
- Kingery, J. N., Roblek, T. L., Suveg, C., Grover, R. L., Sherrill, J. T., & Bergman, R. L. (2006). They're not just little adults: Developmental considerations for implementing cognitive-behavioral therapy with anxious youth. *Journal of Cognitive Psychotherapy*, 20, 263–273. Available from <http://dx.doi.org/10.1891/jcop.20.3.263>.
- Lochman, J. E., Wells, K. C., & Lenhart, L. A. (2008). *Coping power program*. New York, NY: Oxford University Press.
- Marker, C. D., Comer, J. S., Abramova, V., & Kendall, P. C. (2013). The reciprocal relationship between alliance and symptom improvement across the treatment of childhood anxiety. *Journal of Clinical Child and Adolescent Psychology*, 42, 22–33.
- Masia Warner, C., Colognori, D., Brice, C., Herzig, K., Mufson, L., Lynch, C., . . . Ryan, J. (2016). Can school counselors deliver cognitive-behavioral treatment for social anxiety effectively? A randomized controlled trial. *Journal of Child Psychology and Psychiatry*, 1229–1238.
- McHugh, R. K., & Barlow, D. H. (2010). The dissemination and implementation of evidence-based psychological treatments: A review of current efforts. *American Psychologist*, 65, 73–84.
- McLeod, B. D. (2011). Relationship of the alliance with outcomes in youth psychotherapy: A meta-analysis. *Clinical Psychology Review*, 31, 603–616.
- Merikangas, K. R., He, J. P., Brody, D., Fisher, P. W., Bourdon, K., & Koretz, D. S. (2010). Prevalence and treatment of mental disorders among US children in the 2001–2004 NHANES. *Pediatrics*, 125, 75–81.
- Mohr, D. C., Cuijpers, P., & Lehman, K. (2011). Supportive accountability: A model for providing human support to enhance adherence to eHealth interventions. *Journal of Medical Internet Research*, 13(1), e30.
- Mowrer, O. H. (1939). A stimulus-response analysis of anxiety and its role as a reinforcing agent. *Psychology Review*, 46, 553–565.
- Myers, K., & Comer, J. S. (2016). The case for telemental health for improving the accessibility and quality of children's mental health services. *Journal of Child and Adolescent Psychopharmacology*, 26, 186–191.

- National Advisory Committee on Rural Health. (2002). A targeted look at the rural health care safety net: A report to the secretary, U.S. Department of Health and Human Services. Washington, DC: National Advisory Committee on Rural Health.
- Nelson, E., Barnard, M., & Cain, S. (2003). Treating childhood depression over videoconferencing. *Telemedicine Journal and e-Health*, 9, 49–55. Available from <http://dx.doi.org/10.1089/153056203763317648>.
- Olfson, M., & Marcus, S. C. (2010). National trends in outpatient psychotherapy. *American Journal of Psychiatry*, 167, 1456–1463. doi:10.1176/appi.ajp.2010.10040570.
- Ollendick, T. H., Öst, L. G., Rueterskiöld, L., Costa, N., Cederlund, R., Sirbu, C., . . . Jarrett, M. A. (2009). One-session treatment of specific phobia in youth: A randomized clinical trial in the United States and Sweden. *Journal of Consulting and Clinical Psychology*, 77, 54–57. doi: 10.1037/a0015158.
- Pavlov, I. P. (1927). *Conditioned reflexes*. London: Oxford University Press.
- Pelham, W. E., & Fabiano, G. A. (2008). Evidence-based psychosocial treatments for attention-deficit/hyperactivity disorder. *Journal of Clinical Child & Adolescent Psychology*, 37, 184–214.
- Power, M. J., & Dalgleish, T. (2008). *Cognition and emotion: From order to disorder* (2nd Ed.). New York: Psychology Press.
- Pramana, G., Parmanto, B., Kendall, P. C., & Silk, J. S. (2014). The SmartCAT: An m-health platform for ecological momentary intervention in child anxiety treatment. *Telemedicine and e-Health*, 20, 419–427. Available from <http://dx.doi.org/10.1089/tmj.2013.0214>.
- Puliafico, A. C., Comer, J. S., & Albano, A. M. (2013). Coaching approach behavior and leading by modeling: Rationale, principles, and a session-by-session description of the CALM Program for early childhood anxiety. *Cognitive and Behavioral Practice*, 20, 517–528.
- Reinecke, M. A., & Ginsburg, G. S. (2008). Cognitive-behavioral treatment of depression during childhood and adolescence. In J. R. Z. Abela, & B. L. Hankin (Eds.), *Handbook of depression in children and adolescents* (pp. 179–206). New York: Guilford.
- Samoilov, A., & Goldfried, M. R. (2000). Role of emotion in cognitive-behavior therapy. *Clinical Psychology: Science and Practice*, 7, 373–385.
- Sandler, I. N., Ostrom, A., Bitner, M. J., Ayers, T. S., & Wolchik, S. (2005). Developing effective prevention services for the real world: A prevention service development model. *American Journal of Community Psychology*, 35, 127–142.
- Santucci, L. C., & Ehrenreich-May, J. (2013). A randomized controlled trial of the Child Anxiety Multi-day Program (CAMP) for separation anxiety disorder. *Child Psychiatry and Human Development*, 44, 439–451.
- Scheeringa, M. S., Weems, C. F., Cohen, J. A., Amaya-Jackson, L., & Guthrie, D. (2011). Trauma-focused cognitive-behavioral therapy for posttraumatic stress disorder in three-through six year-old children: a randomized clinical trial. *Journal of Child Psychology and Psychiatry*, 52, 853–860.
- Shirk, S. R., Karver, M. S., & Brown, R. (2011). The alliance in child and adolescent psychotherapy. *Psychotherapy*, 48, 17–24.
- Sibley, M. H., Pelham, W. E., Jr, Derefinko, K. J., Kuriyan, A. B., Sanchez, F., & Graziano, P. A. (2013). A pilot trial of supporting teens' academic needs daily (STAND): A parent-adolescent collaborative intervention for ADHD. *Journal of Psychopathology and Behavioral Assessment*, 35(4), 436–449.
- Sibley, M. H., Smith, B. H., Evans, S. W., Pelham, W. E., & Gnagy, E. M. (2012). Treatment response to an intensive summer treatment program for adolescents with ADHD. *Journal of Attention Disorders*, 16, 443–448. doi: 10.1177/1087054711433424
- Skinner, B. F. (1953). *Science and human behavior*. New York: Macmillan.
- Southam-Gerow, M., & Kendall, P. C. (2000). Cognitive-behaviour therapy with youth: Advances, challenges, and future directions. *Clinical Psychology and Psychotherapy*, 7, 343–366. Available from [http://dx.doi.org.ezproxy.fiu.edu/10.1002/1099-0879\(200011\)7:5](http://dx.doi.org.ezproxy.fiu.edu/10.1002/1099-0879(200011)7:5).

- Storch, E. A., Geffken, G. R., Merlo, L. J., Mann, G., Duke, D., Munson, M., . . . Goodman, W. K. (2007). Family-based cognitive-behavioral therapy for pediatric obsessive-compulsive disorder: Comparison of intensive and weekly approaches. *Journal of the American Academy of Child and Adolescent Psychiatry*, 26, 469–478. doi: 10.1097/chi.0b013e31803062e7.
- Suveg, C., Kendall, P. C., Comer, J. S., & Robin, J. A. (2006). Emotion-focused cognitive-behavioral therapy for anxious youth: A multiple-baseline evaluation. *Journal of Contemporary Psychotherapy*, 36, 77–85.
- Watson, J. B., & Raynor, R. (1920). Conditioned emotional reactions. *Journal of Experimental Psychology*, 3, 1–14.
- Weersing, V. R., Rozenman, M. S., Maher-Bridge, M., & Campo, J. V. (2012). Anxiety, depression, and somatic distress: Developing a transdiagnostic internalizing toolbox for pediatric practice. *Cognitive and Behavioral Practice*, 19, 68–82. Available from <http://dx.doi.org/10.1016/j.cbpra.2011.06.002>.
- Weisz, J. R., Chorpita, B. F., Palinkas, L. A., Schoenwald, S. K., Miranda, J., Bearman, S. K., . . . Gibbons, R. D. (2012). Testing standard and modular designs for psychotherapy with youth depression, anxiety, and conduct problems: A randomized effectiveness trial. *Archives of General Psychiatry*, 69, 274–282. doi: 10.1001/archgenpsychiatry.2011.147
- Weisz, J. R., Sandler, I. N., Durlak, J. A., & Anton, B. S. (2005). Promoting and protecting youth mental health through evidence-based prevention and treatment. *American Psychologist*, 60, 628–648. Available from <http://dx.doi.org/10.1037/0003-066X.60.6.628>.
- Weisz, J. R., & Weiss, B. (1989). Cognitive mediators of the outcome of psychotherapy with children. *Advances in Clinical Child Psychology*, 12, 27–51.
- Weisz, J. R., Weiss, B., Han, S., Granger, D. A., & Morton, T. (1995). Effects of psychotherapy with children and adolescents revisited: A meta-analysis of treatment outcome studies. *Psychological Bulletin*, 117, 450–468.
- Whiteside, S. P. H. (2016). Mobile device-based applications for childhood anxiety disorders. *Journal of Child and Adolescent Psychopharmacology*, 26, 246–251.
- Wilamowska, Z. A., Thompson-Hollands, J., Fairholme, C. P., Ellard, K. K., Farchione, T. J., & Barlow, D. H. (2010). Conceptual background, development, and preliminary data from the unified protocol for transdiagnostic treatment of emotional disorders. *Depression and Anxiety*, 27, 882–890. doi: 10.1002/da.20735.
- Wolpe, J. (1952). Experimental neuroses as learned behavior. *British Journal of Psychology*, 43, 243–268.